

S.No	Details
Week 1:	Introduction to DevOps Understanding DevOps principles and practices • History and evolution of DevOps • DevOps culture, collaboration, and communication • Introduction to version control systems (e.g., Git)
Week 2:	Continuous Integration (CI) Overview of CI/CD pipelines • Setting up a CI environment using Jenkins or GitLab CI • Building and testing automation • Integration with version control systems
Week 3:	Assessment-1 Continuous Deployment (CD) Introduction to continuous deployment and delivery • Deployment automation using tools like Ansible or Puppet • Configuration management principles • Blue-green deployments and canary releases
Week 4:	Assessment-2 Containerization with Docker • Introduction to containers and Docker • Building, managing, and deploying Docker containers • Docker networking and storage • Docker Compose for multi-container applications
Week 5:	Container Orchestration with Kubernetes • Understanding Kubernetes architecture • Deploying applications on Kubernetes clusters • Managing Kubernetes resources (pods, deployments, services) • Scaling and autoscaling applications on Kubernetes Assessment-3
Week 6:	Infrastructure as Code (IaC) • Introduction to Infrastructure as Code concepts • Using tools like Terraform or CloudFormation for infrastructure provisioning • Managing infrastructure state and configuration drift • Versioning and collaborating on infrastructure code
Week 7:	Monitoring and Logging • Introduction to monitoring and observability • Setting up monitoring with tools like Prometheus or Nagios • Logging best practices and tools like ELK stack (Elasticsearch, Logstash, Kibana) • Alerting and incident response in DevOps
Week 8:	Advanced Topics and Project Work • Advanced CI/CD pipelines with Jenkins or GitLab CI

- Security best practices in DevOps
- Performance optimization and testing in CI/CD pipelines
- Group project: Design and implement a DevOps pipeline for a sample application

Final Assessment